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Drill Scenarios

Four exercises built from the Hazira gas terminal vapour release and fire, 24 September 2020. Draft learnings for ONGC Fire Services from a ProbeMatrix four-level human-factors investigation.

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Table of Contents

Drill 1. Unignited vapour cloud at night: the 53 minutes before ignition

Drill 2. Major fire with degraded fire water

Drill 3. Ignited fire, escalation and Level-III emergency

Drill 4. Tabletop: the first ten minutes of a gas release

Evidence and limits

For installation Chief Fire Officers and drill planners. Four exercises: three field drills and one tabletop. Each can be printed and run on its own.

Learning purposes: not a statutory document. Derived from a ProbeMatrix investigation export; learnings are evidence-linked but should be validated by the organisation before adoption.

How to use this pack. Each drill opens with **what happened at Hazira** in the phase it rehearses, so the crew knows why the drill exists. It then sets out the objectives, set-up, a timed inject list, the decisions being tested, and a **time-standards table**. The table shows the interval observed at Hazira as a baseline, and leaves a blank for **your installation's target**. Take that target from your ERDMP, the applicable OISD standard or your internal standard. Eyekyam does not propose target times. Adapt equipment names, valves and locations to your own plant before use.

Before every drill: synchronise the umpire's watch with the DCS clock and the fire control room clock, and note the difference. At Hazira, control-room clocks differed by 2 to 12 minutes [EVD-5532]. Without a common clock, the timings below cannot be compared.

The incident in brief. At 02:12:50 a gasket on a recently repaired 1986 condensate meter at the Gas Terminal metering skid began to leak. CISF perimeter guards saw and heard the vapour cloud at 02:15 and informed the Fire Station among others. The cloud was too dense for anyone to reach the leak on foot. The isolation valves could only be closed by hand at the valve, and the emergency plan gave no isolation criteria. The first valve was closed at 03:01. The cloud ignited at 03:08:15, probably on a damaged flameproof light fitting. A Level-III emergency was declared, active cooling began at about 03:50, the fire was out by about 06:00, and all-clear came at 07:30. No one inside the plant was injured. The full account is in *02 Fire Services Learnings, Part A*.

Running the drills well

- Drill the system, not the people. Debriefs ask "what made this hard?", not "who was slow?".
- Record every timed decision point in the drill record, with actions and owners (see learning FS-2).
- Run Drill 1 and Drill 4 at night at least once a year. The Hazira event began at 02:12.

Drill 1. Unignited vapour cloud at night: the 53 minutes before ignition

Type: field drill, night, with the plant control room and CISF taking part. **Duration:** about 60 minutes plus debrief. **Learnings tested:** FS-3 (pre-ignition phase), FS-4 (logging), IF-1 (isolation criteria), IF-2 (reachable isolation).

What happened at Hazira

At 02:15 a watch-tower guard heard a sharp hiss and saw a vapour cloud rising behind the Gas Terminal. The guard called the CISF main gate, **the Fire Station** and the LPG control room [EVD-4286]. Within a minute the CCTV showed the cloud spreading across the road and the piping area [EVD-9941]. Two engineers from the Gas Terminal control room tried to reach the leak on foot and were driven back by the density and the smell [EVD-1460]. The shift in-charge team arrived at the gate at about 02:35, could not enter either, and **told the Fire Station to stop all vehicle movement in the plant** [EVD-9612]. Senior staff were called one after another. The emergency plan had no criterion for isolation and no delegated authority [EVD-4052]. The first valve, MOV 54, was closed by hand at 03:01. By 03:00 the vapour was "all around even at height" [EVD-8213]. The cloud ignited at 03:08:15, **four minutes after the upstream valves were closed** [EVD-4507]. The record does not describe what the fire service did in this window [GAP].

Objectives

1. Receive and act on a gas-release report from security before any fire exists.
2. Complete ignition-source control (vehicles, hot work, the agreed electrical isolations) without anyone entering the cloud.
3. Stage crews and appliances upwind, outside the cloud envelope, on routes that do not cross it.
4. Make and record a water-curtain / dispersion decision.
5. Get from the control room, and log, the isolation status: what is closed, what can be closed, and from where.
6. Account for everyone in the affected area and set a cordon.

Set-up

- **Location:** your highest-inventory skid near an occupied building (the Hazira equivalent is a metering skid beside a control room).
- **Conditions:** night, light wind (state the direction at the briefing), and a visible cloud simulated by smoke or marked by flags and cones at defined radii.
- **Participants:** fire station shift, fire control room, the plant control room for the area, the shift in-charge, CISF watch tower and CCTV room.
- **Rule:** no one enters the marked cloud area. Any attempt is recorded as an observation, not a failure.

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Injects

T+ (min)	Inject	Delivered to
0	Watch-tower guard reports "loud hissing, white cloud rising behind [unit]" by intercom	Fire Station, CISF main gate, one other control room (**not** the controlling control room)
3	CCTV room confirms "heavy leakage" on the monitor	Controlling control room
5	Gas detection annunciator alarms in the controlling control room	Control room
8	Control-room engineers report they cannot approach: "visibility nil, smell overpowering"	Shift in-charge, Fire Station if asked
15	Wind shifts 45 degrees towards the fire station access road	Fire officer on scene
20	Shift in-charge arrives at the gate and cannot enter. Asks the Fire Station for advice	Fire officer
30	Control room: "the isolation valve is local only, inside the cloud"	Fire officer, if they asked for isolation status; otherwise held
45	Umpire: "vapour visible at height across the whole area"	All
50	Umpire calls ENDEX before ignition, or runs on into Drill 3	

Decisions being tested

- Does the fire service treat this as **its** phase, or wait for a fire?
- Who orders the vehicle stop and the hot-work stop, and how fast? (At Hazira, about 20 minutes after the first call.)
- Is there an advance agreement on which electrical circuits may be isolated, and from where?
- Is a water curtain available and used without entering the cloud?
- Does the fire officer **ask** for isolation status, and does the answer change the tactics?
- Does anyone propose an isolation the plan does not authorise, and what happens then?

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Time standards

Interval	Hazira baseline	Your target	Drill result
Security report to Fire Station receipt	~0 min (Fire Station called at 02:15)		
Fire Station receipt to turnout	*not recorded*		
Receipt to ignition-control checklist complete (vehicles, hot work, electrics)	~20 min to the vehicle-movement halt (02:15 to ~02:35); rest not recorded		
Receipt to crews staged upwind outside the cloud	*not recorded*		
Receipt to water-curtain decision recorded	*not recorded*		
Receipt to isolation status obtained by the fire officer	*not recorded* (first isolation at 03:01, 46 min after detection)		
Receipt to head-count of the affected area complete	*not recorded*		

Debrief questions

1. At what point did this become our incident, not someone else's?
2. What did we need to know from the control room, and how long did it take to find out?
3. If the cloud had ignited at T+50, where would our crews and appliances have been?

Drill 2. Major fire with degraded fire water

Type: field drill with live pump operation. **Duration:** about 45 minutes plus debrief. **Learnings tested:** FS-1 (diesel pump availability), FS-4 (logging).

What happened at Hazira

During the fire, fire-water tank levels and header pressure **stayed healthy throughout**. Three electric fire-water pumps (47PE003 A, C and D) carried the cooling load [EVD-2432]. But of the four 410 m³/h diesel fire-water pumps, **two were under maintenance, one ran for 30 minutes, and the fourth trips after two minutes**. The reason for the tripping had not been identified, and no corrective action had been recorded [EVD-2432]. Grid power stayed available all night. If it had failed, the backup that exists for exactly that case was largely not there. The fire damaged electrical fittings within about 15 m of the skid.

Objectives

1. Know the true availability of every fire-water pump at turnout.
2. Keep header pressure up when electric pumps are lost.

3. Recognise and handle a diesel pump that starts and then trips.
4. Request and connect alternative water (mutual aid, portable pumps) in time.
5. Declare a fire-water impairment and its compensating measures.

Set-up

- **Scenario:** a major fire at a process skid needs cooling of the skid and the adjacent equipment.
- **Pre-drill state (umpire's sealed card):** two diesel pumps marked "under maintenance". This can be real or simulated, but use the actual state if your pumps are degraded.
- **Participants:** fire station shift, pump house, electrical control, fire control room, mutual-aid partner (by phone, or live if arranged).
- **Safety:** pump trips are simulated by the umpire telling the pump house. No protective device is defeated.

Injects

T+ (min)	Inject	Delivered to
0	Fire reported at [skid]; cooling of adjacent equipment needed	Fire Station
5	Electric fire pumps running; header pressure normal	Pump house
12	**Grid power lost** to the fire-water electric pumps (fire damage to the supply route)	Electrical control, pump house
13	Diesel pump 1 starts	Pump house
15	**Diesel pump 1 trips** (simulated) after 2 minutes. Cause unknown	Pump house
16	Header pressure falling (umpire gives values)	Fire control room
20	Umpire asks: "How long can you hold cooling on the current supply?"	Fire officer
30	Mutual-aid tender available at the gate, if requested	Fire officer
40	ENDEX	

Decisions being tested

- Did the fire officer know the true pump availability **before** T+12?
- How quickly was the diesel trip noticed, and by whom?
- Were cooling priorities re-ranked when water became limited?
- When was mutual-aid water requested, and was it requested before it was needed?
- Was a fire-water impairment declared, and what compensating measures followed?

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Time standards

Interval	Hazira baseline	Your target	Drill result
Diesel pumps available at turnout (of installed)	1 of 4 usable; 2 under maintenance; 1 tripping [EVD-2432]		
Loss of electric pumps to diesel pump on line	not tested at Hazira (grid stayed available)		
Diesel pump trip to detection by the pump house / fire control room	not tested		
Header pressure kept within the normal band for the drill	Maintained by electric pumps throughout		
Mutual-aid water requested (T+)	Mutual aid called at ~03:10 (with the Level-III declaration); purpose and arrival not recorded		
Impairment declared after capacity drops below design	No impairment practice recorded		

Debrief questions

1. Before today, did we know how many diesel pumps would really run for the full duration?
2. Who decides that degraded fire water is an impairment, and what changes when it is declared?
3. What is the open action on every pump that failed or was simulated as failing today?

Drill 3. Ignited fire, escalation and Level-III emergency

Type: field drill with the Emergency Control Room activated. **Duration:** about 90 minutes plus debrief. Can follow straight on from Drill 1. **Learnings tested:** FS-4 (logging), IF-5 (situational view), command and handover.

What happened at Hazira

At 03:08:15 the cloud ignited. A major fire engulfed the metering skid and the surrounding piping [EVD-4507]. **Seven seconds later the only CCTV camera's cable burned through**, and the response lost its view of the area [EVD-2752]. At about 03:10 the Site Incident Controller, with senior plant management, declared a **Level-III off-site emergency** and called mutual aid. Process trains and the power and steam units were shut down; grid power stayed available. At about 03:15 the Chief Incident Controller took command in the Emergency Control Room. Isolation and depressurisation of connected units began, and mobile teams closed the trunk-line sectionalising valves at remote stations (confirmed at about 03:30). **Active fire-cooling began at about 03:50**, about 42 minutes after ignition. The record does not explain the interval [GAP]. The fire was extinguished at about 06:00, the last inlet valve was closed when the area could be approached at about 07:00, and all-clear was declared at 07:30 after zero LEL and a full head-count.

Objectives

1. Get water on target and cool the exposed equipment quickly and safely after ignition.
2. Run fire command alongside the Site and Chief Incident Controllers, with a clean handover.
3. Keep a picture of the fire scene when the CCTV is lost.
4. Keep a time-stamped fire-service log from the first call to all-clear.
5. Integrate mutual aid.
6. Complete the head-count and the zero-LEL confirmation before all-clear.

Set-up

- **Scenario:** ignition of a vapour cloud at a process skid. Flames impinge on adjacent piping, and a high-inventory vessel is about 30 m away.
- **Participants:** fire station shift, fire control room, Site Incident Controller, Emergency Control Room, CISF, mutual-aid partner, medical.
- **Role:** a named fire-service **loggist** from the first call.

Injects

T+ (min)	Inject	Delivered to
0	Ignition: "large flash, fire at [skid]"	All
0.5	**CCTV view of the area lost**	Fire control room, ECR
2	Level-III emergency declared; mutual aid called	ECR, fire control room
5	Site Incident Controller role changes hands (senior manager arrives)	Fire officer
10	Chief Incident Controller takes command in the ECR and asks the fire officer for a situation report	Fire officer
15	Flame impingement on the line to the high-inventory vessel	Fire officer
25	Mutual-aid tender at the gate	Fire control room
35	Upstream valves reported closed by mobile teams; depressurisation under way	ECR
60	Fire knocked down; cooling continues	Umpire
75	Area approachable; final isolation and zero-LEL check	Fire officer, operations
90	Head-count complete; all-clear request	ECR

Decisions being tested

- What stood between ignition and water on target, and could any of it be prepared in advance?
- Did the fire officer's priorities (cooling of the high-inventory vessel versus fire attack) match the escalation risk?
- Did the incident controller handovers (T+5 and T+10) lose any information the fire service needed?
- How did the fire officer see the scene without CCTV?
- Did the loggist capture every decision with a time?

Time standards

Interval	Hazira baseline	Your target	Drill result
Ignition to Level-III declaration	~2 min (03:08:15 to ~03:10)		
Ignition to first water on target	*not recorded*		
Ignition to active cooling of exposed equipment	~42 min (03:08:15 to ~03:50)		
Mutual-aid call to arrival at the scene	call ~03:10; arrival *not recorded*		
Ignition to fire extinguished	~2 h 52 min (to ~06:00)		
Fire extinguished to zero LEL confirmed	~1 h (to ~07:00)		
Complete fire-service log kept (yes/no)	none in the record		

Debrief questions

1. Where did the minutes go between ignition and cooling, and which ones could be saved by preparation?
2. What did we lose when the camera went, and what would replace it?
3. Could someone rebuild the whole response from our log alone?

Drill 4. Tabletop: the first ten minutes of a gas release

Type: tabletop exercise around a site plan. **Duration:** about 60 minutes. Low cost, and best run before Drill 1. **Learnings tested:** IF-4 (alarm routing), IF-3 (detection), IF-1 (who decides), FS-4 (clocks).

What happened at Hazira

The leak began at 02:12:50 (CCTV). At 02:15 **two CISF posts saw it at once**. The watch tower informed the CISF main gate, the Fire Station and the LPG control room. The Gas Terminal gate informed the CCTV control room and the CISF main gate [EVD-4286; EVD-6234]. **Neither called the Gas Terminal control room**, which controlled the leaking equipment. The CCTV room verified the leak on its monitor and called that control room at 02:18 [EVD-8871]. The gas detection alarm sounded there at about 02:20. It showed only on a local annunciator, was not recorded in the DCS, and was not linked to shutdown or suppression [EVD-8657; EVD-3452]. The shift in-charge team was called at about 02:35. The DCS clocks across the control rooms disagreed by 2 to 12 minutes, which made the sequence hard to rebuild afterwards [EVD-5532].

Objectives

1. Map who would learn of a release, by which route, and when, at your installation.
2. Find the steps where a report waits for verification or relay.
3. Agree a "one call reaches all" arrangement for a security-detected gas release.
4. Confirm whether the fire control room receives gas-detection alarms directly.
5. Confirm who may isolate without seeking permission, under the ERDMP.

Set-up

- **Materials:** a large site plan, a cloud overlay drawn on acetate, coloured cards for each control point (watch towers, gates, CCTV room, fire control room, each process control room, shift in-charge, Emergency Control Room), and a wall clock.
- **Participants:** fire officers, CISF supervisors, control-room supervisors, shift in-charges, the ERDMP owner.
- **Method:** the facilitator reads the injects. Each participant moves their card and states, aloud and with a time, whom they call and what they say. A scribe draws the call map.

Injects

T+ (min)	Inject
0	Leak begins at [skid]. No one has noticed yet.
2	Watch tower and gate both hear the hiss and see a cloud. Each acts as their procedure says.
3	CCTV room receives one call. What does it do next?
5	Gas detector alarms. Where is it seen: annunciator, DCS, fire control room?
6	Control room tries to reach the leak and cannot. Whom does it call?
8	Who now has the authority to isolate? Under which ERDMP clause?
10	Freeze. The scribe shows the call map. Count the calls and the minutes until the controlling control room, the fire station and the person who may isolate **all** know.

Decisions being tested

- How many calls before the controlling control room knows, and before the fire station knows?
- Is any step a verification hop that delays the report?
- Does a security guard know which control room controls which area?
- Does anyone below senior management have standing authority to isolate?

Time standards

Interval	Hazira baseline	Your target	Tabletop result
Leak onset to first human detection	~2 min (02:12:50 to 02:15)		
First detection to Fire Station informed	~0 min (02:15)		
First detection to controlling control room informed	3–5 min (02:18 to 02:20), via the CCTV room		
Leak onset to gas-detection alarm in the controlling control room	~7 min (to ~02:20)		
First detection to shift in-charge informed	~20 min (~02:35)		
First detection to someone with authority to isolate deciding	~46 min to first isolation (03:01)		
Clock difference between control rooms	2–12 min [EVD-5532]		

Debrief questions

1. Draw the call map we would want. How many calls does it take?
2. Which one change would get the right people informed fastest?
3. What would we write into the ERDMP after today?

Evidence and limits

All Hazira facts come from the investigation narrative (the initial incident report) and the evidence items cited (EVD IDs). The supporting findings are in *02 Fire Services Learnings*. Wherever the record does not describe the fire service's own actions, the baseline says *not recorded* rather than guessing. These are learning drills, not a statement of what ONGC's crews did or should have done that night.

See also: PNGRB *Codes of Practices for ERDMP Regulations, 2010* (as amended), on mock drills and exercises, and OISD-STD-116 on fire-water systems. Check the current editions and their applicability.

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